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Textile Digital Drape Definition - TD3: Comparing Digital to Physical Design Considerations

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Textile Digital Drape Definition - TD_3: Comparing Digital to Physical Design Considerations

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Fashion design is, by nature, a very hands-on activity in which the touch and feel of a textile are considered vital to the design of the garment. Yet when designers work with digital fashion design software, they rely on digital textile definitions that are visual instead of tangible textile swatches. In this research, we created a new Textile Digital Drape Definition (TD_3) that is heavily informed by textile drape properties. We compared the TD_3 to traditional tangible textile interaction, via an intense workshop examining eleven textile sample swatches. The digital drape of the TD_3 follows standardized textile science drape methods, and it is created using the measured properties of the physical textile. The participants (N = 24), all digital fashion designers, undertook a three-phase workshop where they 1) observed the TD_3 and indicated attributes and design considerations for each textile, 2) repeated the process after a tangible interaction, and 3) reflected on the visual TD_3 as a tool for determining attributes and design considerations for digital textiles. The results show which textiles benefit the TD_3 and when tangible interaction is needed to better understand, interpret, and refine design considerations. The TD_3 treats drape as an entity that combines its visual and measured mechanical properties. The TD_3 is relevant for researchers, educators, students, and practitioners in fashion and textile design who work with textiles in digital design software. The research introduces a digital textile definition that effectively informs design considerations in digital design with needed standardized implementation.

Keywords: Digital Textile Definitions; digital design considerations; textile drape; Fashion Design

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1. Introduction and background

Fashion design is traditionally a very hands-on activity in which the touch and feel of a textile are considered vital to garment design. When designers work with digital fashion design software, they depend on visual digital textile definitions, which are screen-based abstractions rather than tangible textile swatches (Roth & Ueda, 2023). Digital design is used in fashion design (Conlon, 2024; Särmäkari, 2023) and in design research, as seen in zero-waste design (Rissanen & McQuillan, 2023) or three-dimensional (3D) woven textile-forms or shape change in textiles (Buso et al., 2022; Roth & Ueda, 2023; Voorwinden et al., 2025). In the digital fashion landscape, the material is an abstraction (Ornati, 2023), however, the tactile experience of textiles (e.g., smooth, soft, hairy, rough, slippery) is crucial in design. The textile experience is influenced by several properties, such as fiber type, yarn twist, and structure (e.g., weave type). Together, these properties and characteristics interact within the weave construction (Ojala et al., 2025; Roth & Ueda, 2023), which ranges from plain weaves to textile-forms (Devendorf et al., 2022; Voorwinden et al., 2025).

In research and design practice, the digital screen-based experience needs to adequately represent the physical experience (Kuijpers et al., 2020), allowing designers to achieve the desired value for "aesthetics, comfort, or functionality" (Zhang et al., 2022). Although research has compared physical textiles with their digital simulations (Kuijpers, 2017; Youn et al., 2025), in design, there is often little to no evidence of the relationship between the physical and digital materials used (Roth & Ueda, 2023). Common user interface requires designers to most often adjust sliders to achieve the required look (Conlon, 2024). The slider values are arbitrary and unitless, not directly related to physical textile measurements; a slider-based design process often yields a purely visual representation that is unrelated to the physical textile (Kuijpers et al., 2024). The effect of adjusting the sliders is illustrated in Fig. 1, where A is the original digital textile definition, and B and C are created by adjusting the sliders.

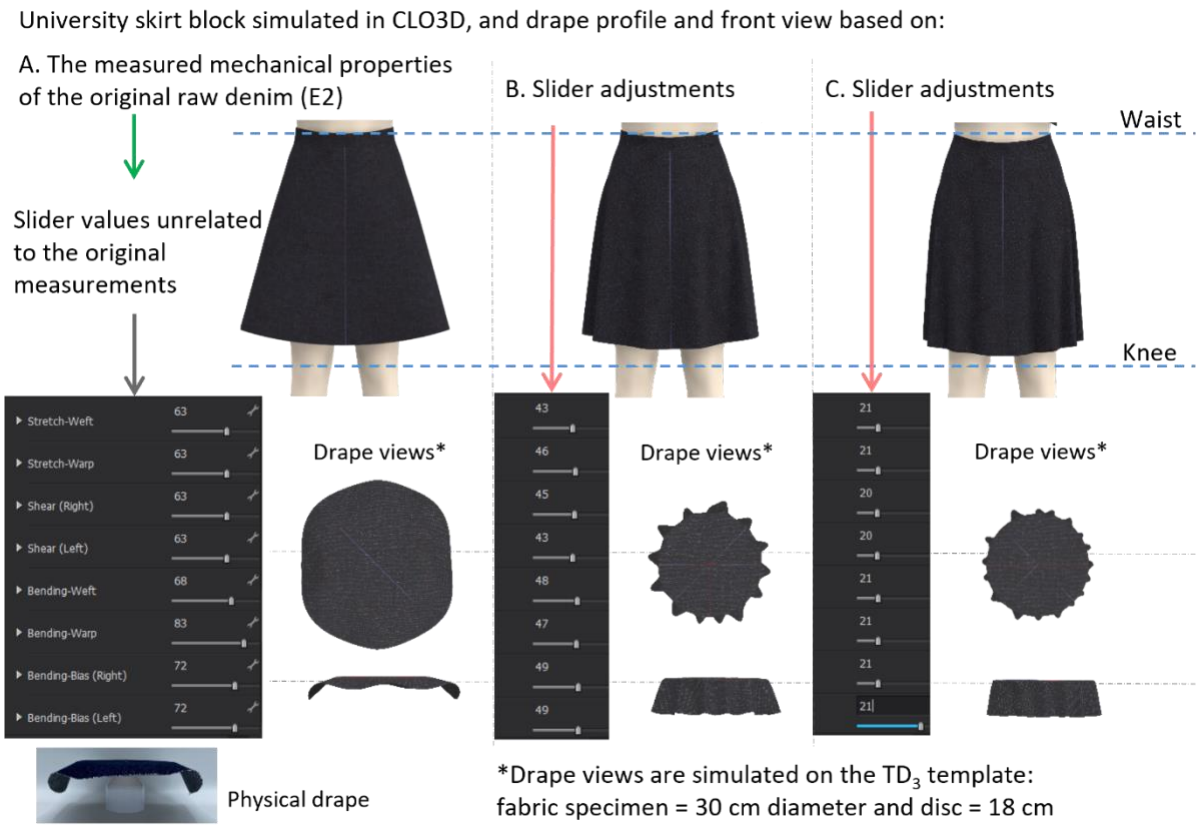


Figure 1. Slider-based design process: sliders unrelated to the textile's mechanical properties, producing digital garments that may be visually convincing but physically unrealizable.

As design technology grows more complex, the relationship between digital processes and physical reality requires continuous recalibration (Goveia da Rocha et al., 2022). The practice of textile design necessitates deploying tacit textile knowledge to generate aesthetically and/or haptically compelling outcomes (Igoe, 2010). Fashion designers develop an understanding of a textile's properties and characteristics through bodily engagement: touching, feeling, and manipulating the material (Petreca et al., 2015). Fashion designers often use tangible interaction to shape design considerations. Yet, textile science-based drape analysis methods allow designers to refine their design considerations (Kuijpers et al., 2024). By using drape-analysis methods, designers can shift their design process toward a "materially oriented design approach" as emphasized by Verbeek (2005), in which "material aesthetics" takes precedence over purely visual aesthetics (p. 232). Material aesthetics in fashion design demands continuous negotiation and exploration to integrate textile material and technological function into a coherent whole (Van Dongen et al., 2019). Material-centered design research is essential for ensuring quality. Wiberg's (2014) materiality lens balances wholeness, material properties, details, and texture as interconnected aspects of a design. Acknowledging material aesthetics in digital fashion design raises the research question: to what extent and in what ways can visually observable and scientifically measurable definitions of digital textiles, including digital drape, inform or transform designers' design considerations that traditionally emerge from tangible interaction?

Drape may hold the key to screen-based textiles (Kuijpers et al., 2024). This requires that at the intersection of fashion design practice and textile science there is a need to disentangle the specific design considerations from measured textile properties and examine how physical properties are

mediated through simulated drape. We hypothesize that an intersection of technical and design understanding is needed to bridge the gap between tangible interaction and screen-based abstraction.

To achieve a visually effective textile definition, this research introduces a Textile Digital Drape Definition (TD_3) and validates it with digital fashion designers. TD_3 is a tool that aims to bridge tangible interaction and screen-based disembodiment. As van Zilt et al. (2022) demonstrate, tools can bridge interdisciplinary gaps and increase domain-specific understanding. While scientific drape is a measure of mechanical behavior, the TD_3 indicates a textile's mechanical drape through its drape shape, drape coefficient, weight, and texture. Inspired by textile science methods for measuring drape (Cusick, 1968), the workshop participants considered the digital drape of a virtual textile, reflecting on the design considerations it elicited. Participants then engaged with the tangible textile used to create the virtual sample and reflected again. The differences between the phases are analyzed to understand what the designer learned in the digital phase versus the tangible phase. This research builds upon the approach and broader challenges described in “Fold, Stand and Drape” (Kuijpers et al., 2024), but, in contrast, it first establishes a digital textile definition and then confirms the findings through tangible interaction.

2. Methodology

2.1. Participants, workshop, and selected textiles

Fashion designers (N = 24) with digital design experience were recruited from the Amsterdam Fashion Institute (AMFI), a fashion design degree program, for a three-hour research workshop. AMFI is internationally oriented, and the students mainly originate from the Netherlands, Germany, and other European countries. Participants, third- and fourth-year BSc students, completed informed consent forms and received Ethics Review Board-approved information forms. Participants had at least two years of experience with 3D design software, including avatar customization and pattern creation, and most had more than two years of experience in physical fashion design. They were previously encouraged to work in a hybrid manner, exploring the interaction between digital and physical designs, by making physical samples.

In the workshop, participants recorded design considerations for eleven textiles across three phases: digital drape observation, tangible interaction, and overall reflection (Fig. 2). Participants were informed of the three phases but not the specific activities. Prior to each phase, a presentation and introduction with instructions were given (see electronic supplement (ES) https:// DOI.org/10.21942/uva.32144512). The presentations focused in phase 1 on drape measurement research and, in phase 2, on material-oriented design research, using examples from Wiberg (2014), Giaccardi & Karana (2015), van Kesteren et al. (2007), and visual materials perception (Fleming, 2014). In each phase, participants completed phase-specific digital questionnaires that included 5-point scales with a ‘cannot observe’ option to avoid forced responses, as well as open-ended questions. To maintain anonymity and enable linking responses across various survey sections, participants received a unique random ID on a physical card. This ID was known only to the participants and was entered during each survey stage.

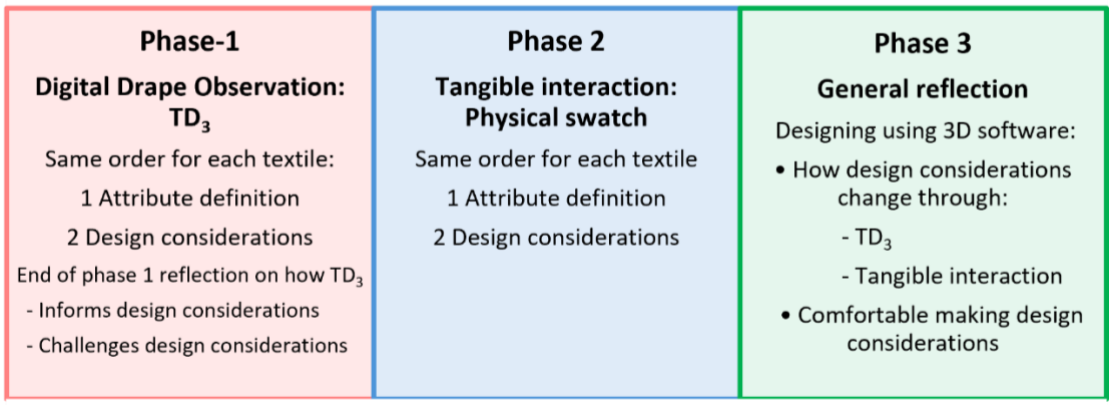


Figure 2. Workshop phases.

The workshop methodology adapts the "Fold, Stand, and Drape" approach by Kuijpers et al. (2024) using their standardized textile library and attribute selection. The textile set (see Table 1) was augmented with a limp-draping fabric (C3) to emphasize differences in drape and with an example fabric (B3) to practice the digital drape phase.

Table 1. Selected textiles.

Textile ID/Code	End-use category	Description	Weight g/m²	Face Drape Coefficient
21F00013/A1	Sheer/special occasion	Silk Organza	21	**92%**
21F00015/A2	Sheer/special occasion	Silk Pongé	36	54%
21F00012/B1	Shirts/blouses/dresses	Cotton satin	112	69%
21F00018/B2	Shirts/blouses/dresses	Organic poplin	120	68%
23F00003/B3	Shirts/blouses/dresses	Oxford cotton	126	60%
21F00035/C1	Sweaters/jumpers	Knit winter sweat	295	64%
21F00036/C2	Sweaters/jumpers	Inlay knit	280	59%
25F00000/C3	Tops/dresses/skirts	Thin, loose knit	257	14%
22F00001/D1	Suiting/coats	Felted wool	279	75%
22F00002/D2	Suiting/coats	Wool blend	198	51%
21F00020/E1	Casual wear/jeans	Heavy Twill	381	95%
22F00000/E2	Casual wear/jeans	Raw Denim	470	83%

During the digital drape phase 1, participants observed the TD_3 (Fig. 3). During tangible interaction phase 2, participants touched rectangular textile swatches measuring 30 x 30 cm (Fig. 4). Participants answered the same questions in both phase 1 and 2; they first determined each textiles' qualities, properties, and characteristics, referred to in this research as attributes. A printout of the terminology prevented misinterpretation of the attributes (ES). Participants indicated design considerations for each textile and its use. Participants reflected on how TD_3 informed their design considerations and indicated which textiles they found most challenging to evaluate. The final phase was a general

reflection in which participants explicitly articulated how the TD_3 changed their design considerations for textiles in a digital environment, and the value of tangible interaction in this context (ES includes the questions).

Each TD_3 includes the textiles' face-drape coefficients (Table 1 and Fig. 3). The drape coefficients were obtained using an upgraded Cusick drape meter for image capture and analysis, enabling objective measurement of drape coefficients, as in the International Organization for Standardization (ISO, 2008). Eighteen measurements (n = 18) per textile were taken using a 30 cm diameter specimen, and the average of the textile's face side (n = 9) was used.

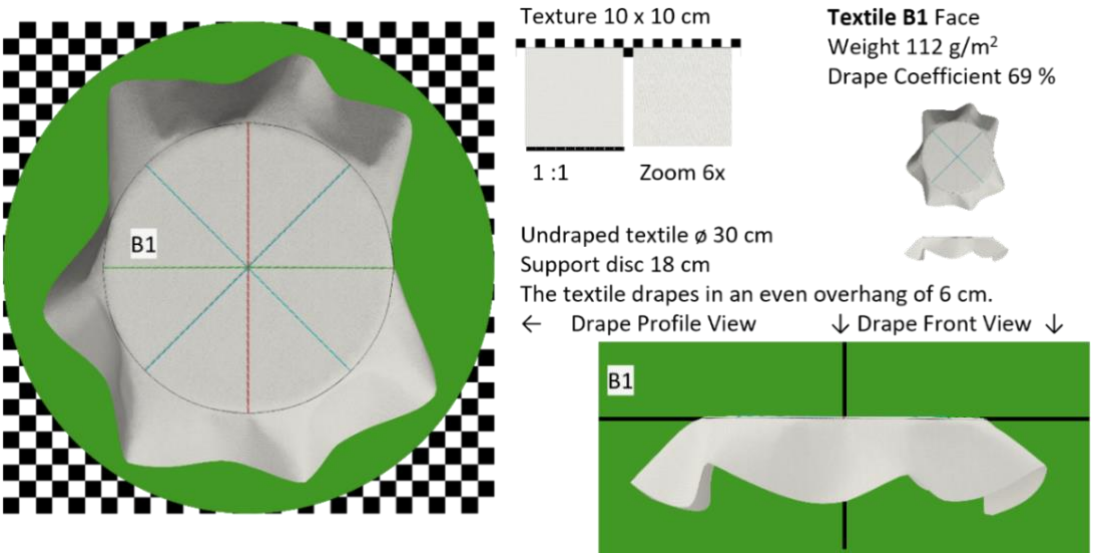


Figure 3. Digital drape observation phase 1: TD_3 of textile B1.



Figure 4: Tangible interaction phase 2.

2.2. Creation of the Textile Digital Drape Definition TD_3

To circumvent the lack of universal testing methods, the proprietary CLO kit was used to measure the textiles' mechanical properties, achieving data compatibility within CLO3D. Cutting and testing three strips in each grain direction (warp, weft, bias) increased the accuracy of bending, shear, and elongation measurement (Kuijpers et al., 2020; Youn et al., 2025). Bending was measured face-up. The textile's face visuals were captured with a Vizoo scanner, and if a different backside (e.g., denim) was present, it was captured as well. The visual files and the averaged property measurements were inserted into the CLO3D system to generate the CLO fabric file (.zfab).

A digital template designed to drape the textile consists of a 30-cm-diameter pattern representing the textile specimen and an 18-cm support disc (Kuijpers, 2017), like the Cusick drape meter. The template (ES) was created in CLO3D. To allow free draping around the support disc, the disc was locked in place and assigned a transparent fabric file to preserve the appearance of semi-transparent textiles. The textile specimen was stitched to the disc edge to maintain centering during simulation (Fig. 5A–B). A top stitch indicated the grainlines: warp (red), weft (green), and bias (blue). For consistency across specimens, the undraped template was positioned in bird's-eye view, zoomed to the upper and lower edges of the software window, and saved as a custom view file (.zcmr) (Fig. 5C–D). The draped front view was created and saved using the same procedure.

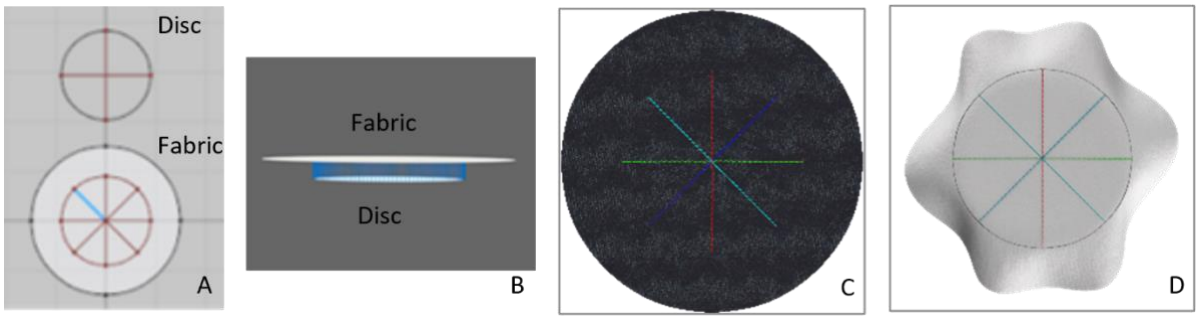


Figure 5. Digital drape template with disc and textile: (A) patterns; (B) Textile stitched on the disc; (C) Textile E2 before draping; (D) Textile C1 after draping.

The digital drape template was saved with the "Fitting (accurate fabric)" settings and a particle distance of 5 mm. To drape each digital textile, its zfab fabric file was dragged onto the template. To refresh the template between simulations, the textile pattern was undraped and correctly re-applied to the disc. For each textile, an image of the simulated drape was rendered at 3543 by 3543 pixels, resulting in 30 x 30 cm renders with the drape centered (Fig. 5C-D). Moreover, a texture render was generated from a hanging rectangle at 1:1 scale and 6x zoom.

The TD_3 included renders of digital textiles' drape and texture, along with weight and drape coefficient values. To avoid influencing participants' design considerations by end-use category, composition, or description, this information was not included in the TD_3. Each TD_3 visual relates the draped textile to the undraped textile and the disc (Fig. 5C-D) and shows their positions within a square composed of 1 x 1 cm squares (Fig. 6-7). The introduction explained how to interpret the TD_3 using Fig. 6-7. Participants answered the questions for each textile based on their observation of the textile-specific TD_3.

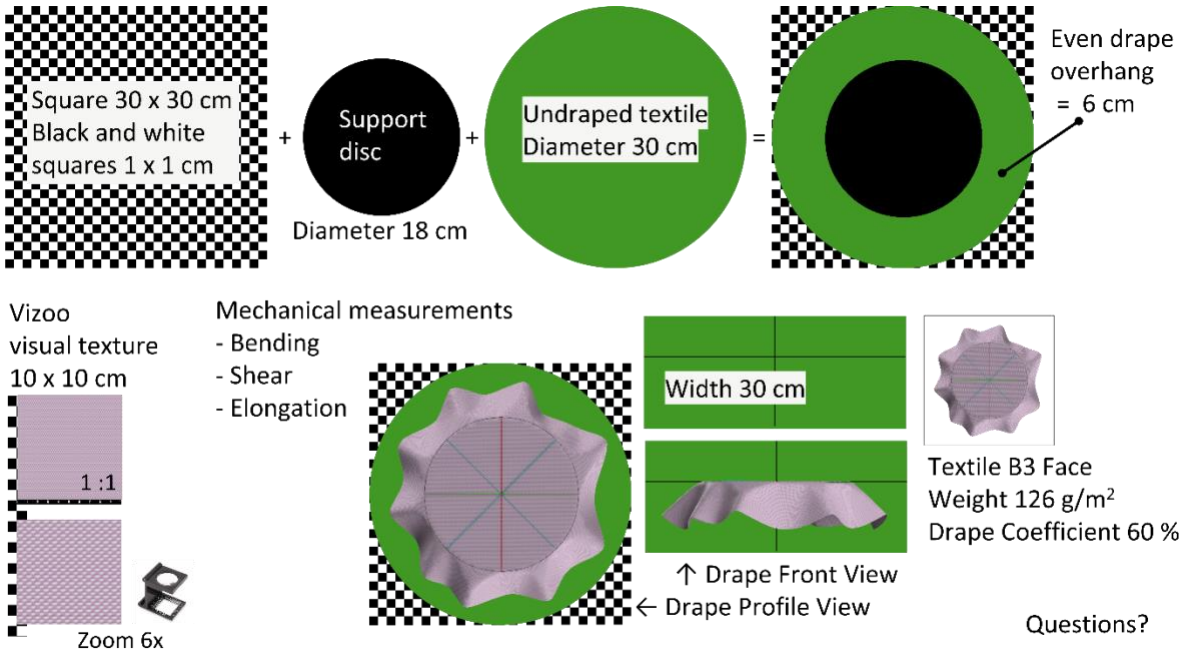


Figure 6. Visualization of how the TD_3 is composed.

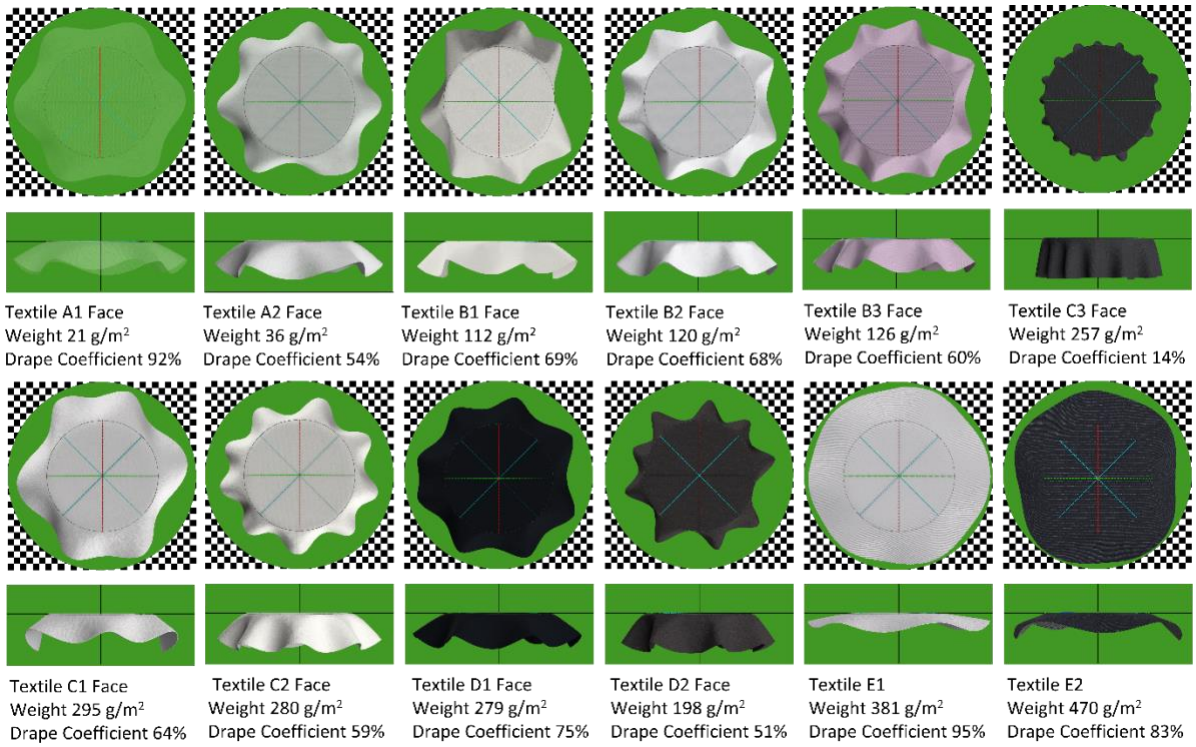


Figure 7. Overview of the digital textile drape.

2.3. Analysis

A dataset was compiled from the combined digital drape observations and participants' tangible interactions. This includes the specified attributes and design considerations reflected by each participant. The dataset was pseudo-anonymized using the randomly assigned ID number.

The dataset was used to map profiles of the 29 recorded attributes and design considerations, showing how responses changed between digital drape observations and tangible textile interactions. The mean score for each attribute, design consideration, and TD_3 element value across all participants was plotted on radar graphs. Participants' mean scores on 29 textile attributes were compared between both phases using Pearson correlation coefficients (*r*) for each attribute, excluding unidentifiable attributes (0), e.g., when participants could not observe stretch or shininess. Textile drape coefficients were then plotted against attribute correlations to assess whether higher drape values corresponded to stronger attribute agreement, and a regression line was fitted after omitting non-significant textiles. Participants' design considerations were hierarchically coded into categories and end-use groups following the thematic networks approach (Attride-Stirling, 2001), with a value of 1 assigned when a consideration was present. Absent, unclear, or uncertain responses were flagged as N/A, unsure, or unspecified. Coded responses were compared with predefined end-use categories (see Table 1) and classified as valid if they aligned with any of three predefined levels (most common, additional possible, or uncommon but plausible), and as incorrect otherwise, for both phases. Valid response percentages per textile and phase were visualized in an overlaid radar chart. Thematic analysis of participants' reflections followed the logic of iterative coding, progressive theme condensation, and verbatim grounding (Attride-Stirling, 2001) and was conducted in Excel using a color-coded scheme (ES). Codes progressed from broad exploratory categories to four organizing themes in phase 1 and three in phase 3, treated separately in Section 3.2 due to their distinct reflective functions.

3. Results and discussion

3.1. Alignment and shifts between digital drape observation vs tangible interaction

Participants' alignment and shifts across the 29 attributes are visualized in radar charts for each textile (Fig. 8), with the correlation coefficient *r* indicating consistency in perception between phases. The *r* value is essential for interpreting cases with lower correlations ($r < .80$), where visual differences in attribute peaks become difficult to discern, revealing perception stability per textile.

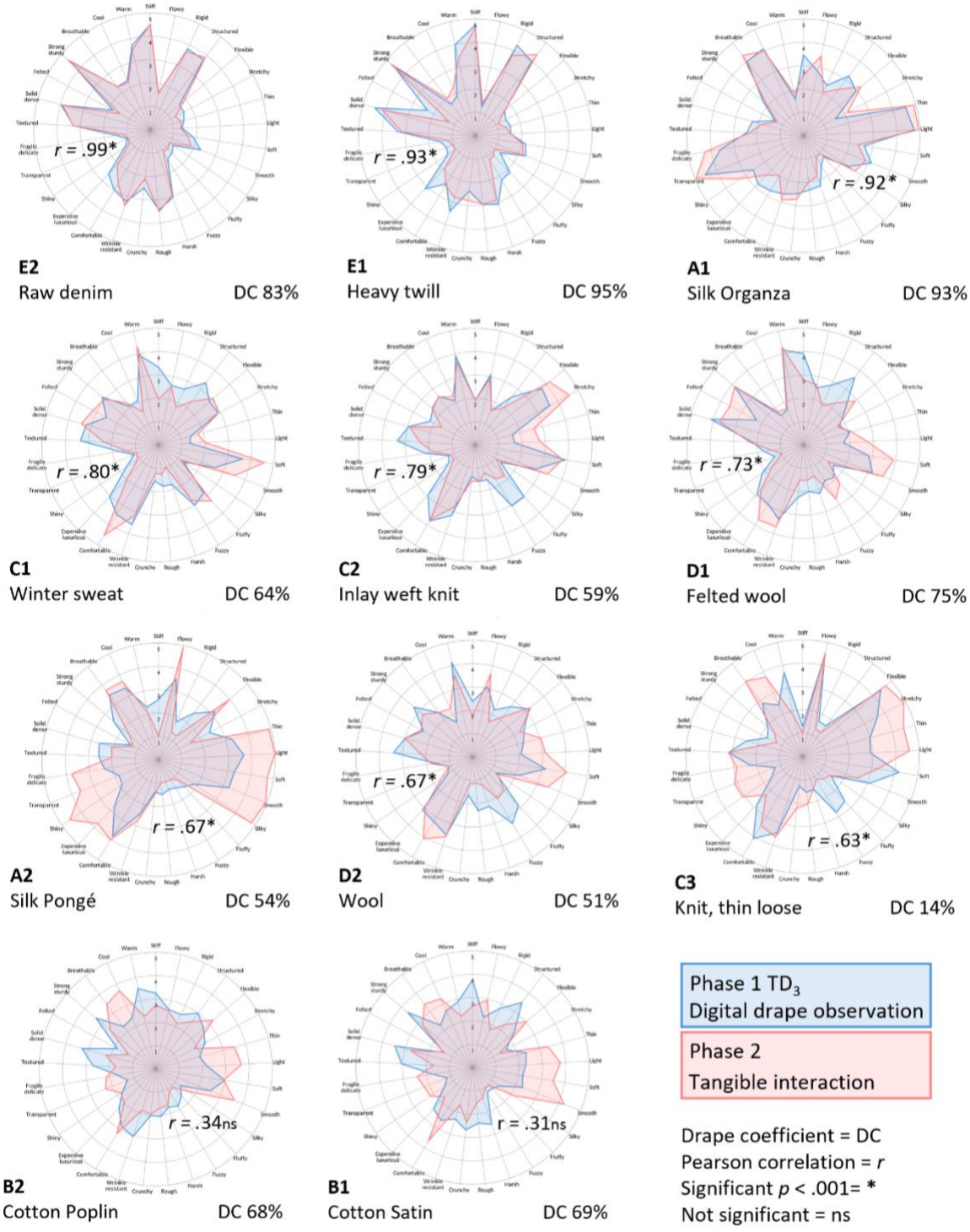


Figure 8. Indicated attributes: Digital drape observation versus tangible interaction.

Participants' attribute understanding shows a significant positive correlation for nine of eleven textiles between phase 1 (digital drape observation) and phase 2 (tangible interaction). These correlations indicate that both methods largely capture the same perceptual character for most textiles. The low *r* values for textiles B1 and B2 indicate poor correlation. Depending on the strength of the correlation (high *r* value), the overall attribute means indicate that the general intensity of perception remains consistent across methods, while specific attributes differ.

Participants' perceptions of stiff and flowy remained stable for E1, E2, C2, and C3 (Fig. 8). For B2, C1, and D1, changes in stiffness did not affect fluidity between phases, suggesting these attributes operate independently for some fabrics. Digital drape observation generally yielded higher stiffness ratings, while tactile interaction produced higher fluidity ratings for A1, A2, B1, and D2. Silk organza was perceived as stiffer in digital drape, aligning with Kuijpers et al. (2024) findings for physical drape. Touch enabled participants to better discriminate transparency, flexibility, and stretchiness. Attribute changes through tactile interaction correlated with touch properties (soft, smooth, silky) across fabrics with good, moderate, and low correlation coefficients (Table 2).

Table 2. Attributes: Phase 1 digital drape observation vs phase 2 tangible interaction.

Correlation Ph1↔Ph2	**Attribute changes (≥ 1 scale)**	**Textile:**
Very high:	***Mainly*:** Slight refinement (A1, E1, E2). ***Decrease*:** fluffy, fuzzy, expensive/luxurious, felted (E1), stiff (A1).	•A1/Silk organza •E2/Raw denim •E1/Heavy twill
Good:	***Decrease*:** stiff, rigid, structured (C1-D1), flexible, fluffy, fuzzy (C2), expensive/luxurious, texture (C1-C2). ***Increase*:** stretchy, thin (C2), light (C1-D1), soft (C1-D1), smooth, comfortable (C1) felted (D1).	•C1/Winter sweat •C2/Inlay knit •D1/Felted wool
Moderate:	***Decrease*:** stiff, structured, stretchy, strong/sturdy (A2). ***Increase*:** flowy, flexible, thin, light, soft, smooth, silky, expensive/luxurious, shiny, transparent, fragile/delicate (A2). D2 and C3 show a similar trend, but specific attributes differ.	•A2/silk pongé •C3/thin, loose knit •D2/wool
Low (ns):	***Mainly*:** considerably flipping, increasing, or decreasing. ***Unchanged*:** felted (B1), flowy and soft (B2).	•B1/cotton satin •B2/cotton poplin

A strong positive correlation exists between the attribute correlation (*r*) and the drape coefficient. Textiles showing strong agreement in attribute rankings between digital drape observations and tangible interactions also tend to exhibit higher DC values. Exceptions are textile B1 and B2, which have relatively high DC (68%–69%) but showed low/non-significant *r* values and were excluded from the correlation analysis. See Fig. 9.

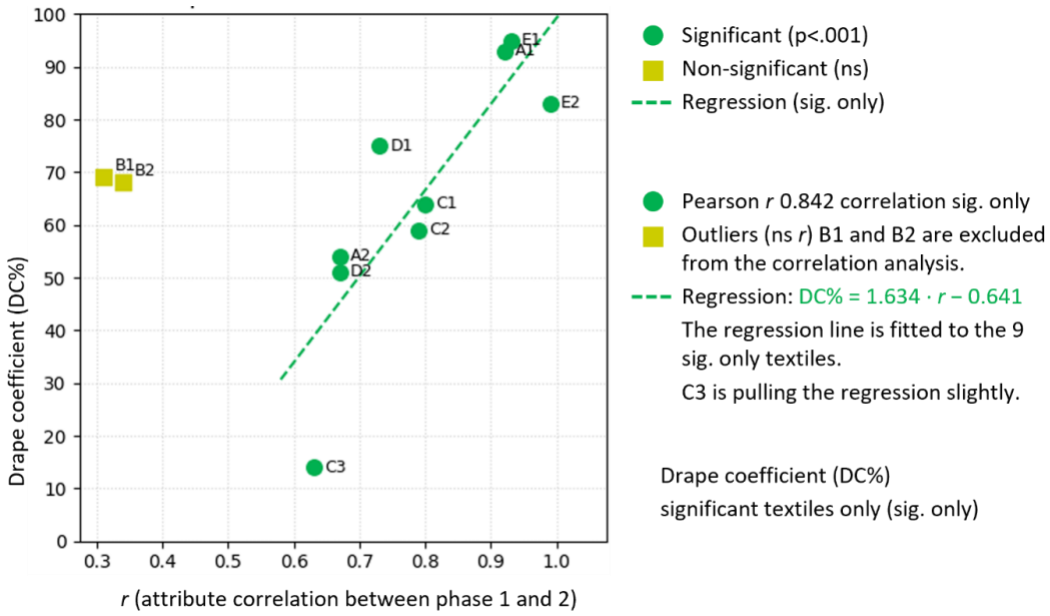


Figure 9: Drape coefficient versus attribute correlation.

Participants' design considerations for each textile were compared to the end-use category (Table 1) or a related suitable end-use. Similarities and differences between phase 1 and 2 are shown in Fig. 10 and Table 3. Six of eleven fabrics showed consistent relationships between digital drape observations and tangible textile interactions. Five textiles prompted refinement and broadening of design considerations. Five textiles initially received unsuitable design considerations or unclear suitability assessments; tangible interaction enabled correction of these assessments.



Figure 10. Design considerations: Digital drape observation versus tangible interaction.

Table 3. Design considerations: Phase 1 digital drape observation vs phase 2 tangible interaction.

Textile	Change %	Design considerations: Changes in P2 tangible interaction
	Consistent: appropriate use in both phases	
E2-raw denim	0%	Highly recognized and indicated in both phases.
	Similar across phases, changes:	
C2-inlay knit	+4%	Slightly broader use.
D2/wool	+4%	Slightly broader use.
A1/silk organza	+8%	Slightly more sheer/special occasion.
C1/winter sweat	+8%	Refining.
C3/thin-loose knit	+13%	Consistent considerations.
	Some unstable & inconsistent in P1, changes:	
B1/cotton satin	+17%	Unclear and unsuitable outerwear considerations, changed to suitable use.
E1/heavy twill	+21%	Unclear responses changed to better recognition of heavy twill/white denim character.
B2/cotton poplin	+21%	Inappropriate considerations, e.g., outerwear and knitwear, changed to appropriate use.
D1/felted wool	+26%	Unclear changed to clearer use/recognition of suiting/coat character.
	Substantial inconsistent in P1, changes:	
A2/silk pongé	+39%	Unsuitable or unclear, changed to appropriate use. Fragile/silk is better recognized.

3.2. Phase 1 digital drape observation reflection on TD_3

Phase 1 reflection revealed how TD_3 informed participants' design considerations and which textiles they found challenging. The methodological approach intentionally excluded metadata from TD_3, leading nine participants to report insufficient information for textile identification. Textile D1 was cited by five participants (two citing missing metadata, three citing unclear visuals). Textiles A1, B1, and B2 were reported three times as difficult to identify without their metadata. Further, missing metadata was noted a few times across various textiles.

Seven participants reported challenges related to preferences or limitations while demonstrating recognition of textile physical properties, suggesting identification through drape observation. Textiles A1 and C3 presented physical (thinness) or digital (transparency, computation time) difficulties. Participants also reported excessive stiffness, softness, or weight in specific textiles. Four participants noted lack of touch; three cited unfamiliarity with the method. Textiles A1, C3, E1, B1, and D1 were most challenging overall. Most participants indicated two or three textiles despite the option to select one (see ES).

Participants' statements on TD_3 affordances were grouped into four organizing themes (Attride-Stirling, 2001), accordingly: Design considerations; experimentation, verification, information; textile understanding; limitations (see Fig. 11).

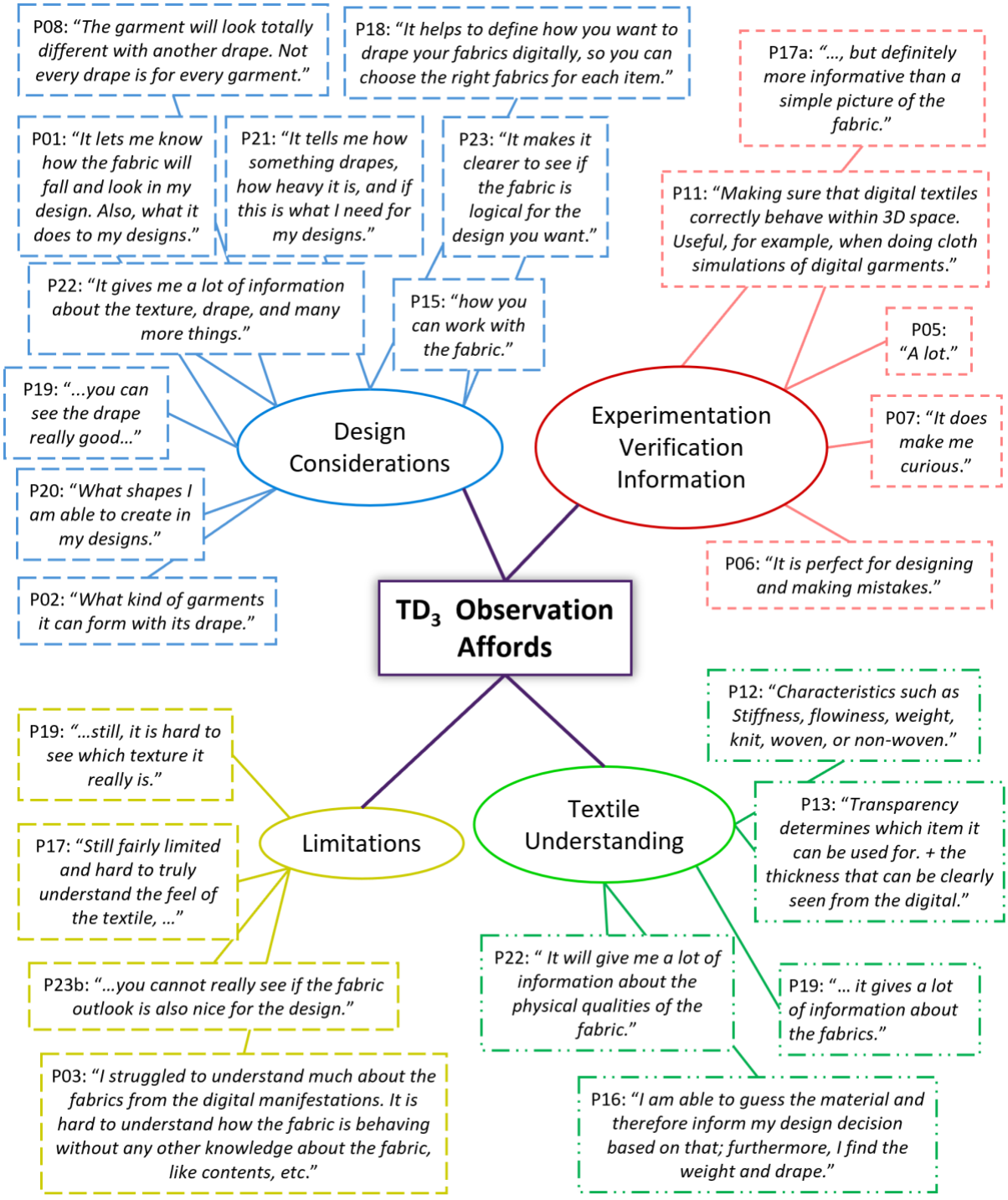


Figure 11. Phase 1 drape observation reflection on TD_3.

The participants clearly stated they needed more information for their design considerations. As P12 noted for textile B2: "*Because the textiles are not named. If it said this is cotton satin and showed the characteristics, it would be easier to consider them based on digital drape.*" This indicates that including the metadata in the TD_3 would enable intrinsic knowledge to yield more targeted design considerations.

3.3. General reflection

Excluding general metadata from the TD_3 forced the participants to read the drape. Although they frequently addressed the missing metadata, their preferences and limitations were closely tied to the textiles' physical properties and thus based on what participants read in the TD_3. Tangible interaction helped participants to confirm, refine, and correct their design considerations. However, the touch partly compensated for insufficient information caused by the excluded metadata in the TD_3. As P17 indicated, "*a lot of the crucial information was gained from analysis of physical swatches.*"

The majority of the twenty participants felt comfortable designing digitally; a few felt partly comfortable; and one felt not at all comfortable. Five participants emphasized the combination with touch, and four highlighted that understanding drape is a competency to develop. Noteworthy is P23's reflection: "*not 100%, but I would give it a chance, and I must say I also have it sometimes with physical material that I misplaced, how it would drape.*" P23 explicitly acknowledges failure based on touch, which supports TD_3 as a good alternative for first design experiments, decisions, and prototyping.

Textile science focuses on measuring mechanical properties (Cusick, 1968; ISO, 2008). Our research shows that the visual structure is equally important. Texture helps interpret drape behavior, which cannot be done solely from the drape shape and coefficient. The drape profiles of B2/organic poplin and D1/felted wool are very similar (Fig. 5), but the textiles differ significantly in weight and end use (Table 1). The drape coefficient of A1, E1, and E2 is relatively similar, but A1 is super light and fragile, while E1 and E2 are sturdy and relatively heavy.

The TD_3, with its visually observable and scientifically measurable digital textile definitions, including digital drape, informs or transforms designers' design considerations. Table 3 shows that TD_3 informs design considerations for half of the textiles in both phases. TD_3 informs and transforms design considerations and material understanding. This is evidenced by the participants' reflections presented in Fig. 12.

The reflections show that participants better understand the potential of textiles; see, for example, the reflections of P07, P08, P20, P21, and P22 (Fig. 12●). How drape informs material understanding and, in parallel, transforms design consideration is clearly stated by P16 and P19 (Fig. 12●●); in this case, the TD_3 mediates "material aesthetics" beyond visual aesthetics as emphasized by Verbeek (2005).

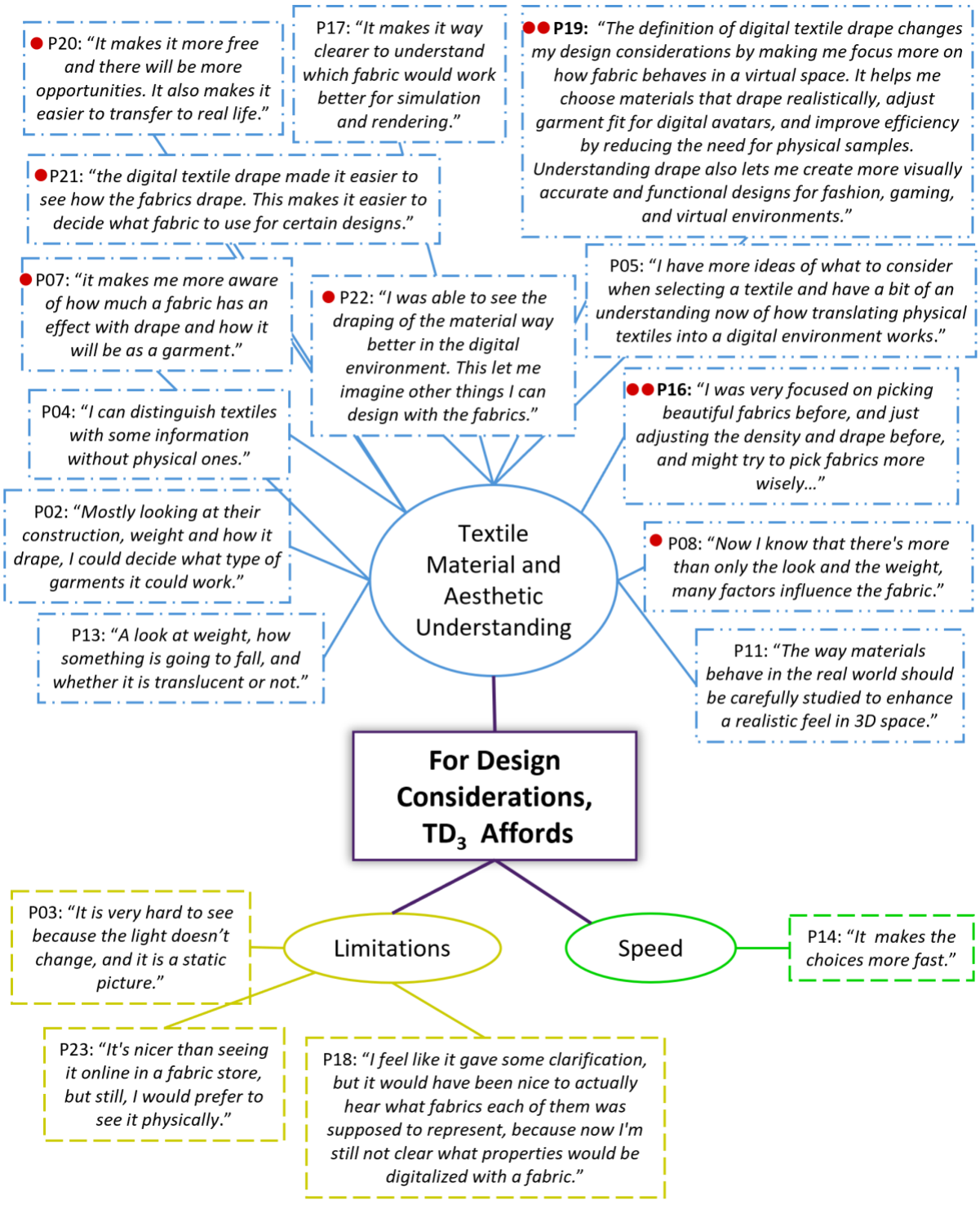


Figure 12. Phase 3 General reflection TD_3.

The value of TD_3 as a tool in digital design is shown in Figure 12. Participants found all elements and indicators in TD_3 important. The average values for Drape Front View, Drape Profile View, and Weight were considered equally important as touch (see Fig. 13). The findings establish that including the ISO (2008) based digital template (Fig. 3 and Section 2.2), the drape views with their related proportions and texture at 1:1 and 6x zoom (Fig. 4) enables researchers and practitioners in the field of fashion and textile design to observe and compare textiles in a digital environment under standardized

conditions. Moreover, they can compare the digital drape with the physical textile drape by using the Fold, Stand, and Drape approach introduced by Kuijpers et al. (2024).

With a reasonably small amount of background in drape, students and practitioners can understand many design considerations through digital design software with TD_3. We demonstrate in this paper that we can drape textiles in a digital environment, and we hope that software developers adopt the TD_3 as a standard and replace meaningless sliders with meaningful measurements.

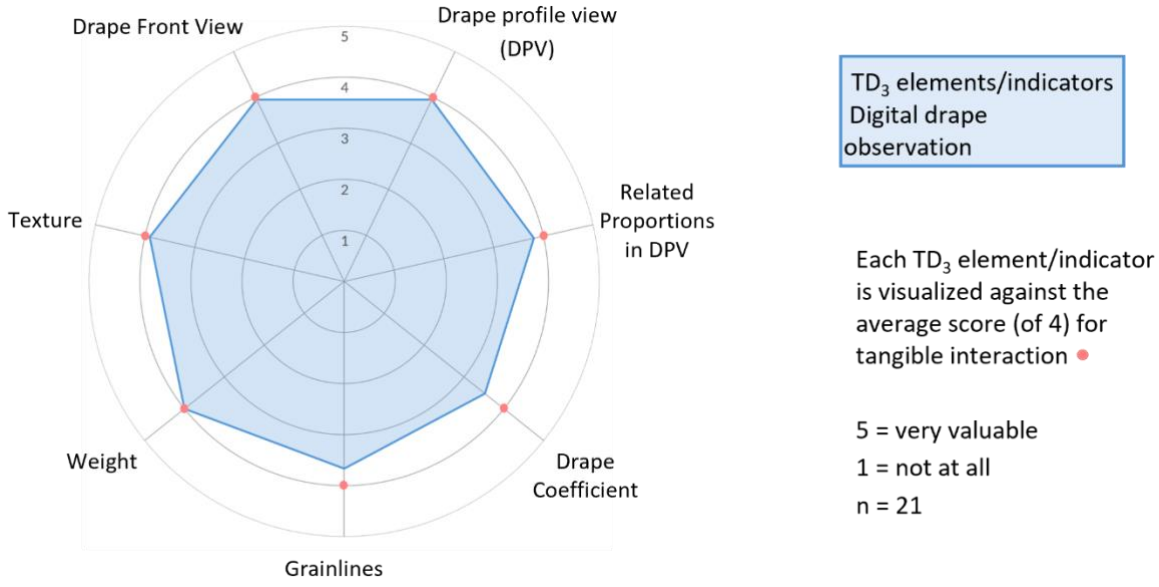


Figure 13. Values TD_3 indicators vs tangible interaction.

The study shows that, despite the absence of metadata, observing TD_3 for six out of eleven textiles provided valuable insights that informed early design decisions aligned with valid end-use considerations. The textiles in the two blue center rings of Fig. 14 can be effectively considered for design using TD_3. The textiles in the two outer rings require tangible interaction to be effectively considered for a design. The textiles marked with a small square in Fig. 14 reveal TD_3's effectiveness in defining attributes. TD_3 identifies attributes with comparable effectiveness across most textiles, with C3 and E1 as exceptions; their attributes require tangible interaction despite TD_3 being sufficient for design consideration.

Drape analysis, as depicted in Fig. 12, enhances understanding of textile materials; however, no direct correlation exists between the drape coefficient and specific design considerations, as exemplified by D2 versus A2 or C1 versus B1/B2 (Fig. 14). Cotton shirting (distinct from denim) typically combines lightweight fabric, a fine weave, and a stiff drape (see Fig. 7 and Table 1), a challenge recognized since the 1960s (Cusick, 1968). During tangible interactions, cotton shirting tends to fold and crease easily. Consequently, a more precise mental understanding of cotton is essential for practitioners and requires careful, deliberate consideration. The valid design considerations for all three knits (C1, C2, C3), the wool blend (D2), and the raw denim (E2) lean on their visually distinctive structures. A1 combines a typical drape, transparency, and weight, making it easy to identify. Understanding the differences between A2 and C2 also requires careful consideration of their drape views, weight, and structure.

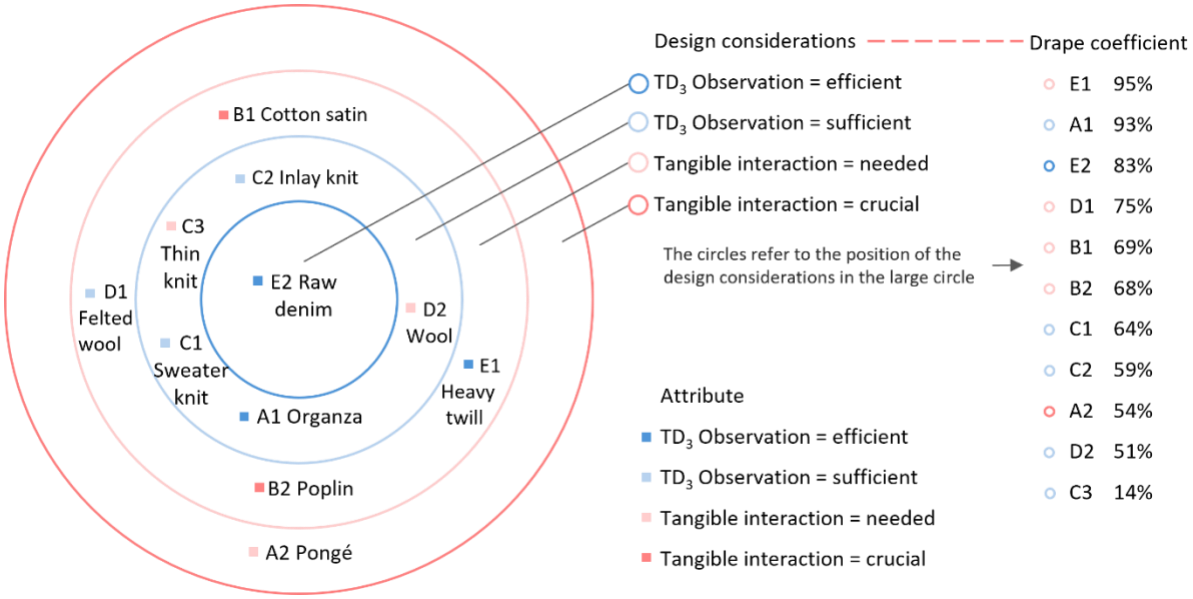


Figure 14. Interpretive framework for TD_3 suitability across textile types, spanning from efficient observation to crucial tangible interaction requirements.

Including textile metadata in the TD_3 enhances confidence and may increase the number of valid end-use considerations. The holistic relationship between drape, weight, structure, attributes (such as material properties and characteristics), and metadata aligns with the wholeness of materiality of the "material lens" (Wiberg, 2014). A more profound understanding of drape opens new possibilities for researchers and practitioners in fashion and textile design by integrating their knowledge of textile end-use relationships with the TD_3. The workshop approach with the materials in ES could be used for educational and self-learning purposes.

4. Conclusions and future perspectives

In this research, designers used a digital drape definition (TD_3) to observe textiles for their design considerations. TD_3 is a tool that indicates a textile's mechanical drape behavior through its drape shape, drape coefficient, weight, and texture. The TD_3 is innovative in digital design practice because it moves textiles beyond simple surface-level digital swatches to include comprehensive, measured material properties. The TD_3 approach isolates drape as an entity, highlighting the interplay between its mechanical and visual properties, which is mostly sufficient in informing textile attributes that lead to design considerations. However, this is not uniform across all textiles, and the research shows which TD_3 definitions are more effective. The results show that the drape is informative for indicating (digital) textiles; this is confirmed by the participants' dissatisfaction with the common yet "missing" generic textile data. Adding the meta-data into TD_3 is important for a deeper understanding of digital drape once mastery of reading drape is achieved. TD_3's works well for several textiles, while others are misinterpreted. Providing designers with a format to understand which textiles they can visually interpret and which they really need to touch is important for their skills development when working with digital 3D software. The research confirms that the TD_3 effectively informs design considerations in digital textile definitions, providing standardized implementation. Designers (and design software programmers) should use the TD_3 prescribed in this work to define textile design attributes and end-use considerations, especially in digital design situations where the tangible textile is unavailable.

More research is needed to refine the understanding of the more difficult-to-interpret textiles. Moreover, the drape coefficient was measured from the physical textile and presented alongside its digital drape. In future work, verification of the alignment between the physical and digital drape should be conducted and incorporated into TD_3. Understanding the similarities and differences between the physical and digital drape will further increase trust and decision-making when working with digital textiles. The impact of adding metadata to design considerations merits further investigation. In tandem, future work should examine how practitioners can be equipped to analyze drape objectively, fostering greater openness to and adoption of drape analysis in design practice.

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Appendix

Supplementary Materials

All supplementary materials related to this article are available on Figshare:

(https://doi.org/[10.21942/uva.32144512]).